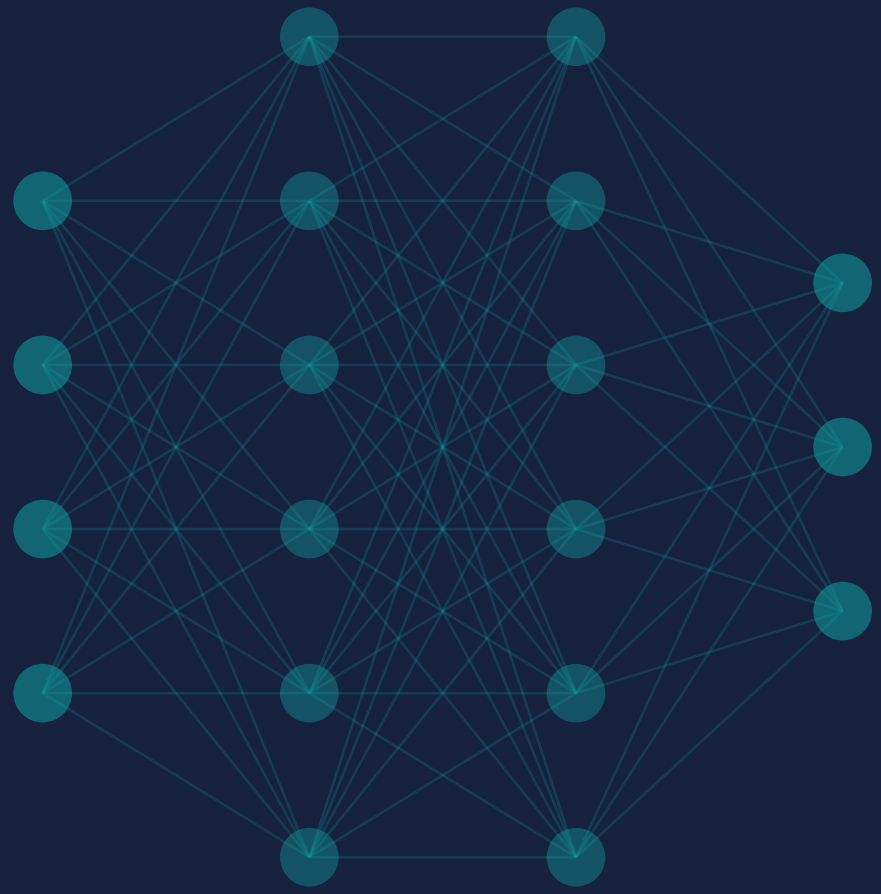
DEEP LEARNING IN MEDICINE

CNNs for Medical Imaging

Teaching a computer to read chest X-rays

A step-by-step project: from raw X-ray images to a trained network that spots pneumonia, and the questions we must ask before trusting it.



LESSON ROADMAP

What we will explore today

1 Why medical imaging?

How doctors use images, and where AI can help.

2 A quick CNN recap

Filters, pooling, and why CNNs are made for pictures.

3 The data

A real, open dataset of children's chest X-rays.

4 Build and train

Prepare the images, design a CNN, and train it.

5 Evaluate like a doctor

Accuracy is not enough. Which mistakes matter?

6 Explain and use wisely

See what the network looks at, and use AI responsibly.

PART 1

Why Medical Imaging?

Doctors look at millions of pictures of the inside of the body. Let us see where a CNN can help.



Doctors use many kinds of medical images

X-ray

Quick and cheap. Shows bones and lungs. Our project uses chest X-rays.

CT scan

Many X-rays taken in a circle and combined into 3D slices of the body.

MRI

Uses strong magnets to show soft tissues, like the brain and muscles.

Ultrasound

Uses sound waves. Famous for showing babies before they are born.

Microscope slides

Zoomed-in pictures of cells and tissue, used to find diseases like cancer.

Eye photos

Pictures of the back of the eye can show signs of diabetes and other illnesses.

Every one of these is a picture, and pictures are exactly what CNNs are good at.

Why can AI help doctors?

1 So many images

Hospitals take huge numbers of scans every day, and each one must be read carefully.

2 Tired eyes

People get tired after hours of reading images. A computer does not get tired.

3 Not enough specialists

Some places have few doctors trained to read X-rays. AI could help them sort urgent cases.

4 A second opinion

AI can point to a spot and say, "Please look here again," so fewer cases are missed.

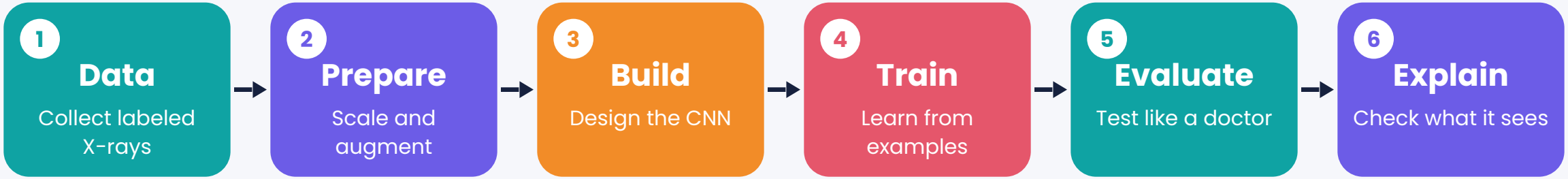
AI is a helper for doctors, not a replacement. The final decision belongs to a trained person.

Our mission today

Build a CNN that looks at a **child's chest X-ray** and answers one question:

Is this X-ray NORMAL, or does it show signs of PNEUMONIA?

We will follow the same 6 steps that real machine learning engineers use:



Each step has a matching section in the notebook, so you can run every part yourself.

This is a learning project, not a real medical tool. Never use it to make health decisions.

What is pneumonia?





NORMAL **PNEUMONIA**

Two real X-rays from our dataset

An infection of the lungs

Germs cause the tiny air sacs in the lungs to fill with fluid. This makes breathing hard.

How it looks on an X-ray

Air looks **dark**. Fluid looks **white and hazy**. Pneumonia often shows up as cloudy white patches.

Why it matters

Pneumonia is a serious illness for young children around the world. Finding it early helps.

Even for experts, the difference can be subtle. That is why this is a good challenge for a CNN.

PART 2

A Quick CNN Recap

Before we build, let us remember how a convolutional neural network sees a picture.



Why use a CNN for X-rays?

1 Neighbors matter

In a picture, nearby pixels belong together. A CNN looks at small patches, not one pixel at a time.

2 One filter, everywhere

The same filter slides across the whole image, so a pattern is found wherever it appears.

3 Learns its own features

Nobody tells it what pneumonia looks like. It discovers useful patterns from examples.

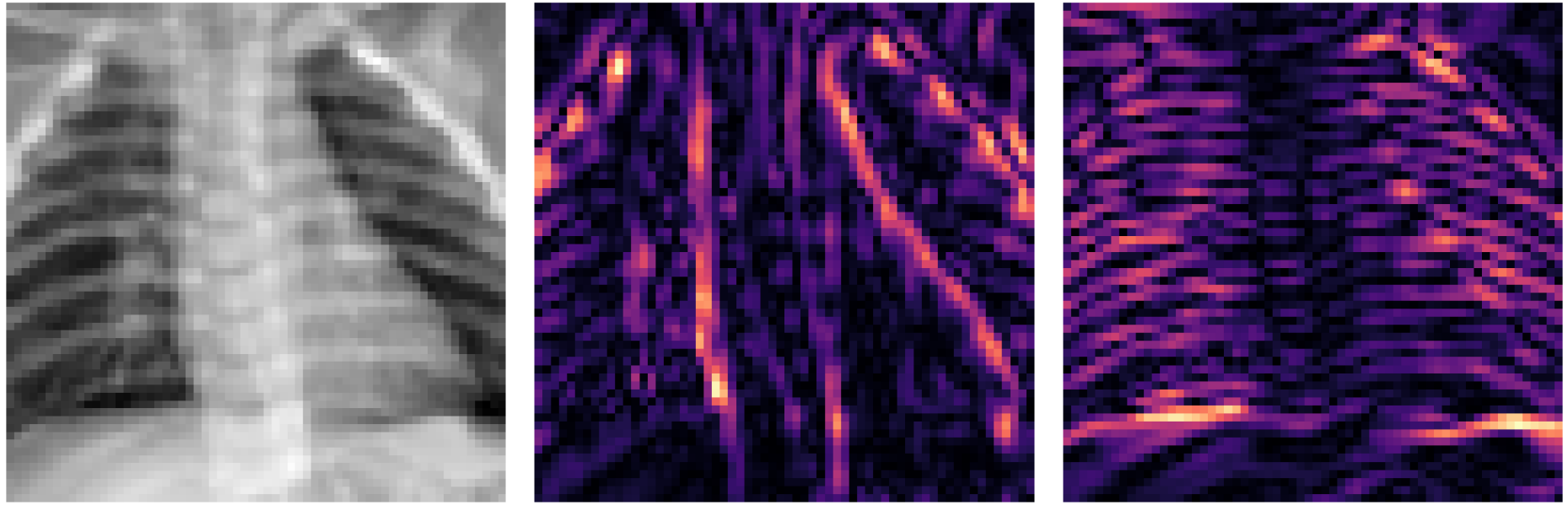
Small layers build big ideas

Edges → Lines and curves → Ribs, heart, lung shapes → Hazy patches → NORMAL or PNEUMONIA

Early layers find simple patterns. Deeper layers combine them into more meaningful ones.

SIMULATION

A filter slides across an X-ray



Vertical-edge filter

1	0	-1
1	0	-1
1	0	-1

What do you see?

The vertical filter lights up the **spine and the sides of the ribs**. The horizontal filter lights up the **ribs**. A CNN learns dozens of filters like these, all by itself.

The building blocks of our CNN

Conv2D

Slides small filters (3 × 3) across the image to find patterns.

ReLU

Keeps positive signals and turns negatives into 0.

MaxPooling

Shrinks the image by half, keeping the strongest signal in each 2 × 2 block.

Flatten

Unrolls the final grid of features into one long list of numbers.

Dense

A regular layer of neurons that combines all the features to decide.

Sigmoid output

One neuron that gives a chance between 0 and 1: "how likely is pneumonia?"

Pattern: Conv → Pool → Conv → Pool … then Flatten → Dense → Output

PART 3

The Data

A network is only as good as the examples it learns from.
Let us meet ours.



PART 3 · THE DATA

Meet the dataset: PneumoniaMNIST

5,856
chest X-rays of children

2
classes: normal and pneumonia

64 × 64
pixels per image (grayscale)

Free
open access, CC BY 4.0 license

Where it comes from

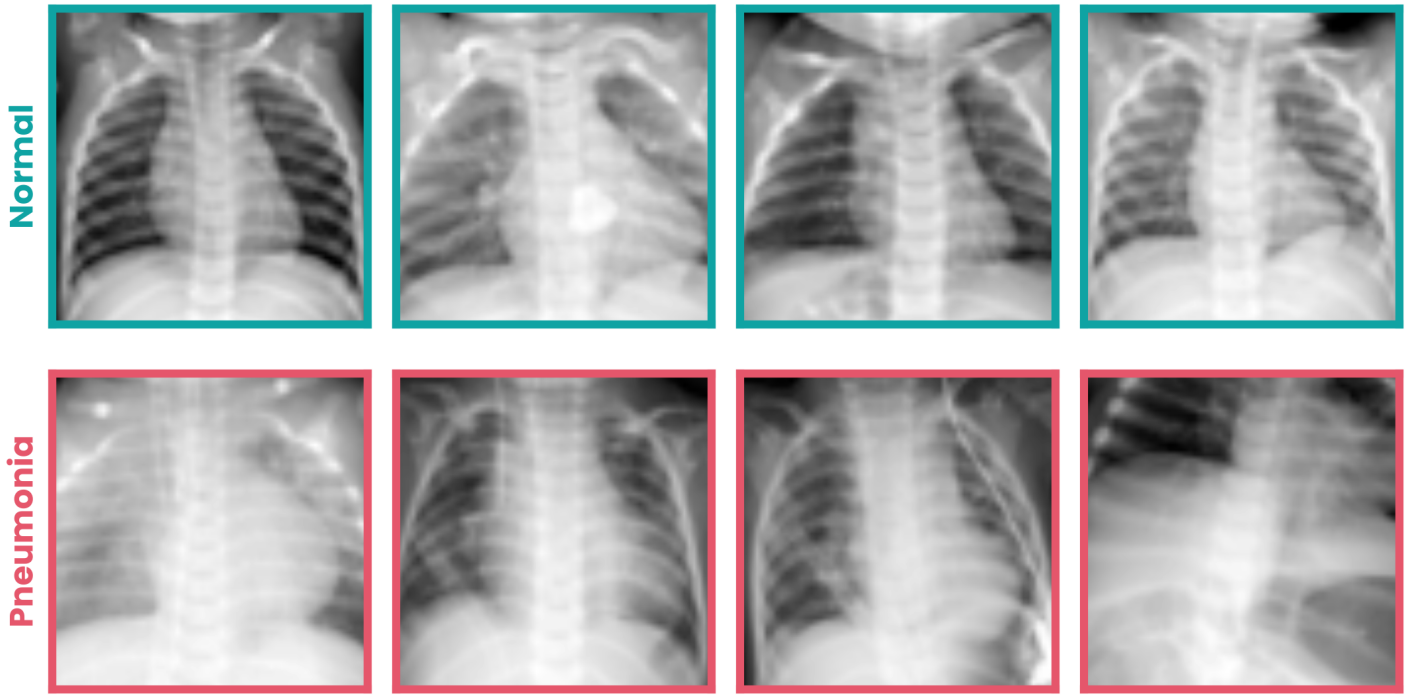
The X-rays were collected at a children's hospital in China and labeled by medical experts. They were published by Kermany and colleagues in 2018, in the journal *Cell*.

Made for learning

MedMNIST is a collection of medical datasets made small and easy to use. Its creators (Yang and colleagues, 2023) shrank the images so they train quickly on any computer.

Download: medmnist.com · Data hosted on Zenodo · Free to use with credit to the authors

Can you spot the difference?



Your turn!

Look at the two rows. What makes the pneumonia X-rays look different from the normal ones?

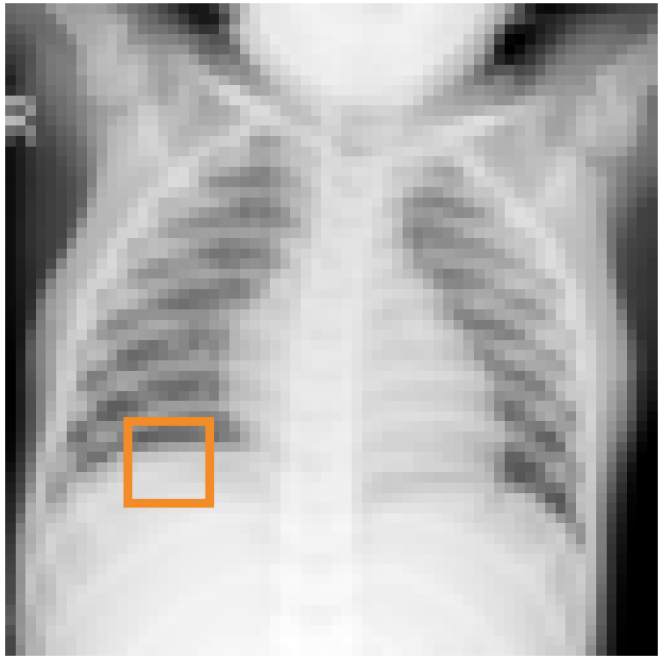
Hint: compare how dark and clear the lungs look.

Not always easy

Some cases are clear. Others are hard, even for people. Small, blurry images make it harder still.

SIMULATION

To the computer, an X-ray is just numbers



The orange box marks an 8 × 8 patch.

→

130	124	111	101	102	104	108	116
96	85	90	95	98	98	103	109
115	121	134	144	148	149	152	153
174	177	178	181	184	186	186	187
190	195	195	196	196	194	192	189
199	201	200	198	197	195	194	193
200	199	199	200	201	202	202	201
198	200	202	203	205	205	206	206

Zoomed-in pixel values

Pictures are grids

Each pixel is one number from **0 (black)** to **255 (white)**.

One 64 × 64 image has **4,096** numbers.

Real hospital X-rays have **millions** of pixels. We use small images so training is fast.

Three piles of X-rays

Training set

4,708 X-rays

Practice problems. The network learns from these.

Normal: 1,214

Pneumonia: 3,494



Validation set

524 X-rays

Practice quizzes. We use them to pick the best epoch and settings.

Normal: 135

Pneumonia: 389



Test set

624 X-rays

The final exam. Used only once, at the very end.

Normal: 234

Pneumonia: 390



The test set must stay hidden until the end. Peeking at it is like seeing the exam before you take it.

SIMULATION

A hidden trap: unbalanced data

Training X-rays



About **3 out of 4** training X-rays show pneumonia.

Meet the "lazy" model

Imagine a model that **always** says "pneumonia," without even looking.

On our test set, it would score **62.5% accuracy**, because 390 of the 624 test X-rays show pneumonia.

But it would send **every healthy child** for treatment they do not need!

Our fix: class weights

We tell the network that mistakes on the smaller group count more. A mistake on a normal X-ray counts **1.94×**, and a mistake on a pneumonia X-ray counts **0.67×**. Now both groups matter equally during training.

PART 4

Build and Train the CNN



Prepare the images, design the network, and let it practice.

Step 2: prepare the images

1 Scale the pixels

Divide every pixel by 255, so values go from 0 to 1. Networks learn best with small numbers.

2 Add a channel

Color photos have 3 channels (red, green, blue). X-rays have 1. So each image becomes 64 × 64 × 1.

3 Keep the splits apart

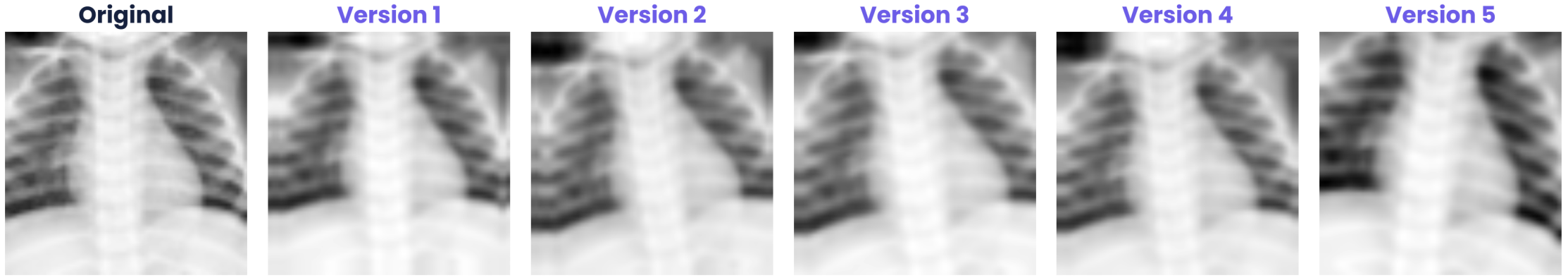
Training, validation, and test X-rays never mix. This keeps our final score honest.

Before		Calculation		After
pixel = 204	→	**204 ÷ 255**	→	**pixel = 0.8**

The picture looks exactly the same. Only the scale of the numbers changes.

SIMULATION

Data augmentation: more variety for free



What changes

Each time the network sees an X-ray, it is **tilted a little** (up to about 11°), **shifted** up to 5%, and **zoomed** up to 10%.

Why it helps

Real X-rays are never taken in exactly the same way. Small changes teach the network to focus on the lungs, not on the exact position.

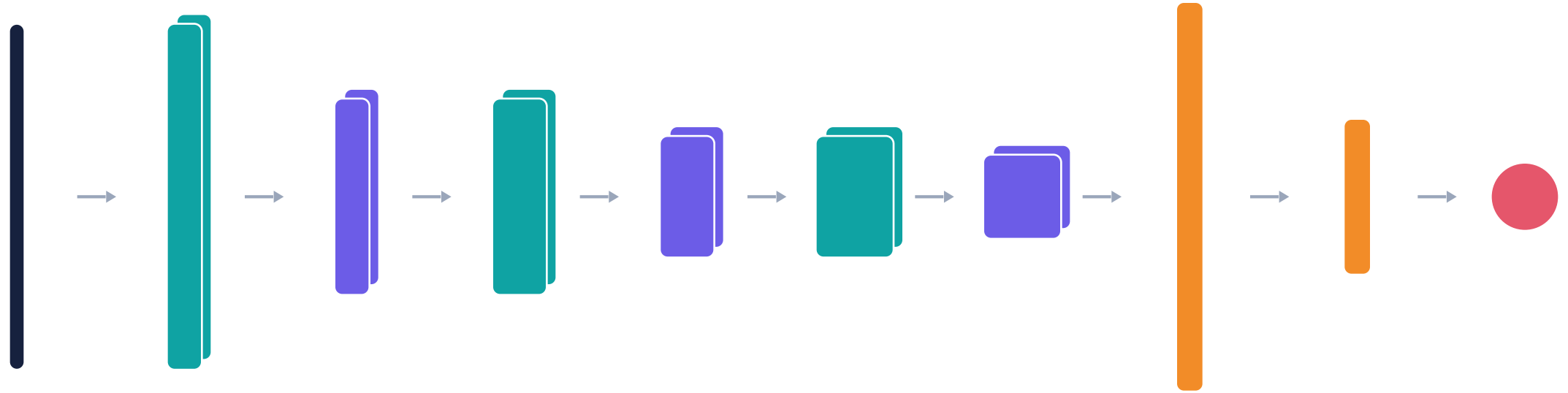
What we do NOT do

We never flip the X-ray left to right. The heart sits on the left side, so a flipped chest would not look real.

Augmentation happens **only during training**. Validation and test X-rays are never changed.

Step 3: our CNN, layer by layer

The picture gets smaller, but the number of feature maps grows.



Input	Conv	Pool	Conv	Pool	Conv	Pool	Flatten	Dense	Output
64×64×1	64×64×16	32×32×16	32×32×32	16×16×32	16×16×64	8×8×64	4,096	32	1

154,433 knobs to tune

Every weight and bias in these layers is learned during training.

Output: one chance

A sigmoid neuron gives a number from 0 to 1: the chance of pneumonia.

Two tools to stop memorizing

Dropout (30%)

During training, 30% of the neurons in one layer are switched off at random, a different 30% each time.

It is like a team practice where random players sit out. Everyone must learn to contribute, so no one player memorizes the answers.

Early stopping

After every epoch, we check the validation score. If it has not improved for 6 epochs in a row, training stops.

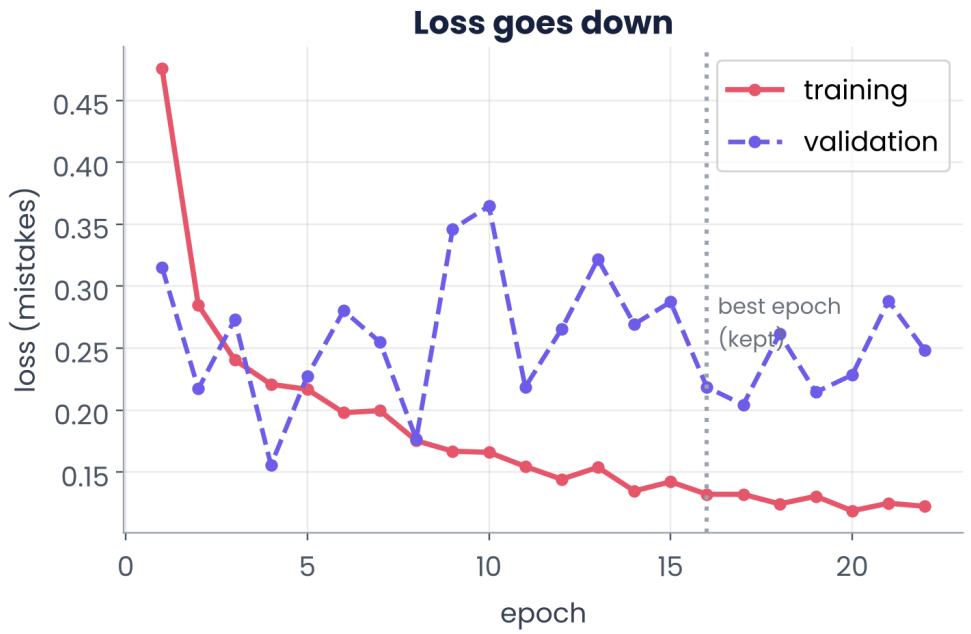
Then we go back to the **best epoch**, like keeping your best practice test instead of your last one.

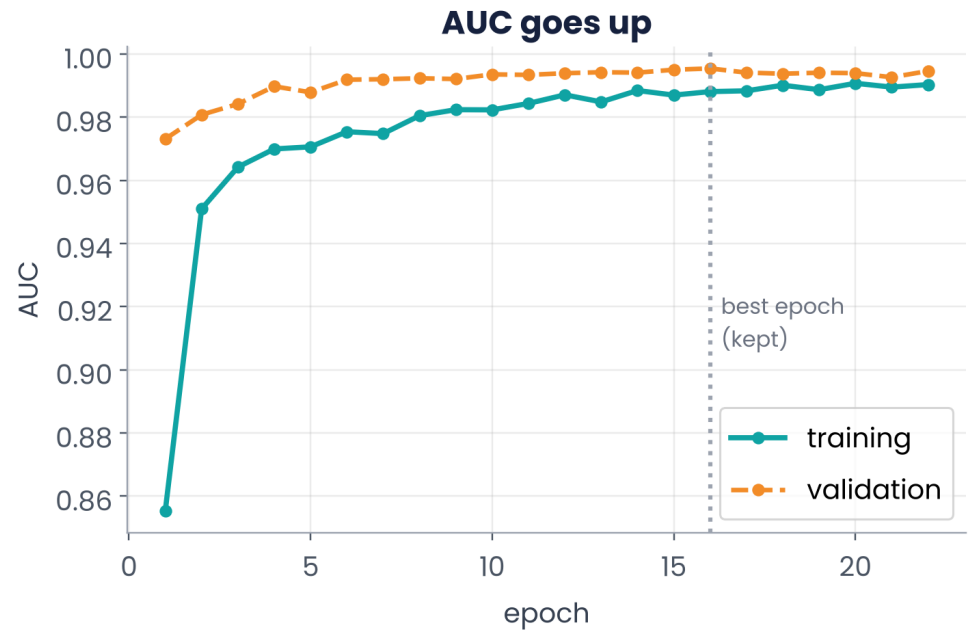
Training settings

Optimizer **Adam**	Loss **Binary cross-entropy**	Batch size **64 X-rays**
Max epochs **25**	Class weights **On**	Best model by **Validation AUC**

SIMULATION

Step 4: watch it train





It learned fast

Training stopped after 22 epochs. The best epoch was **epoch 16**, so we kept that version.

Why is validation bumpy?

The validation set is small (524 X-rays), so a few changes in answers move the score a lot.

Why is validation higher?

During training, dropout and augmentation make the task harder on purpose. Validation is the "easy" version.

PART 5

Evaluate Like a Doctor



In medicine, a single accuracy number is not enough.
We must ask: which mistakes did it make?

PART 5 · EVALUATE LIKE A DOCTOR

Is accuracy enough?

The "lazy" model

62.5%

test accuracy · Always says pneumonia

Our CNN

93.1%

test accuracy · Looks at every X-ray

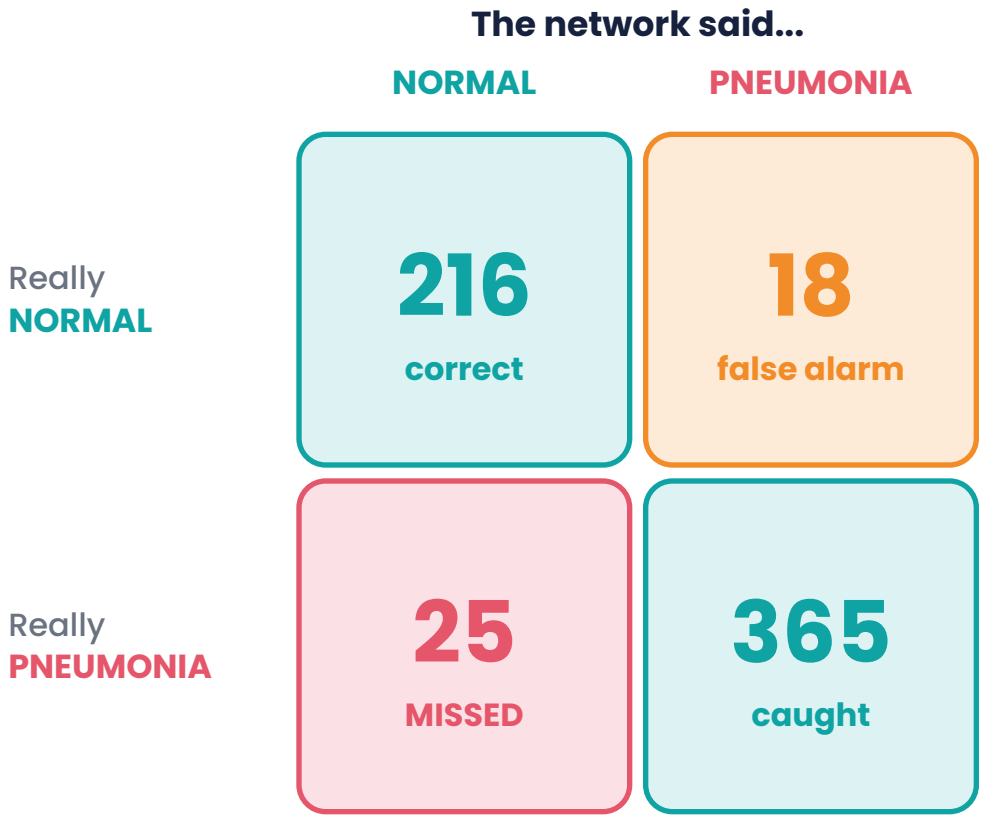
But accuracy hides the most important question

Our CNN got 43 of the 624 test X-rays wrong. Were those **sick children sent home**, or **healthy children sent for more tests**? In medicine, these two mistakes are not equally bad.

To answer this, we use a confusion matrix.

SIMULATION

The confusion matrix



Test set: 624 X-rays · threshold 0.5

Read it like this

Rows show the **truth**. Columns show what the **network said**. The diagonal (top-left to bottom-right) holds the correct answers.

25 missed cases

Children **with** pneumonia that the network called normal. This is the most dangerous mistake.

18 false alarms

Healthy children that the network flagged. A doctor would check them and find nothing wrong.

Two scores doctors care about

Sensitivity

Of all the sick children, how many did we catch?

caught ÷ all sick

365 ÷ (365 + 25)

= 93.6%

Specificity

Of all the healthy children, how many did we clear?

cleared ÷ all healthy

216 ÷ (216 + 18)

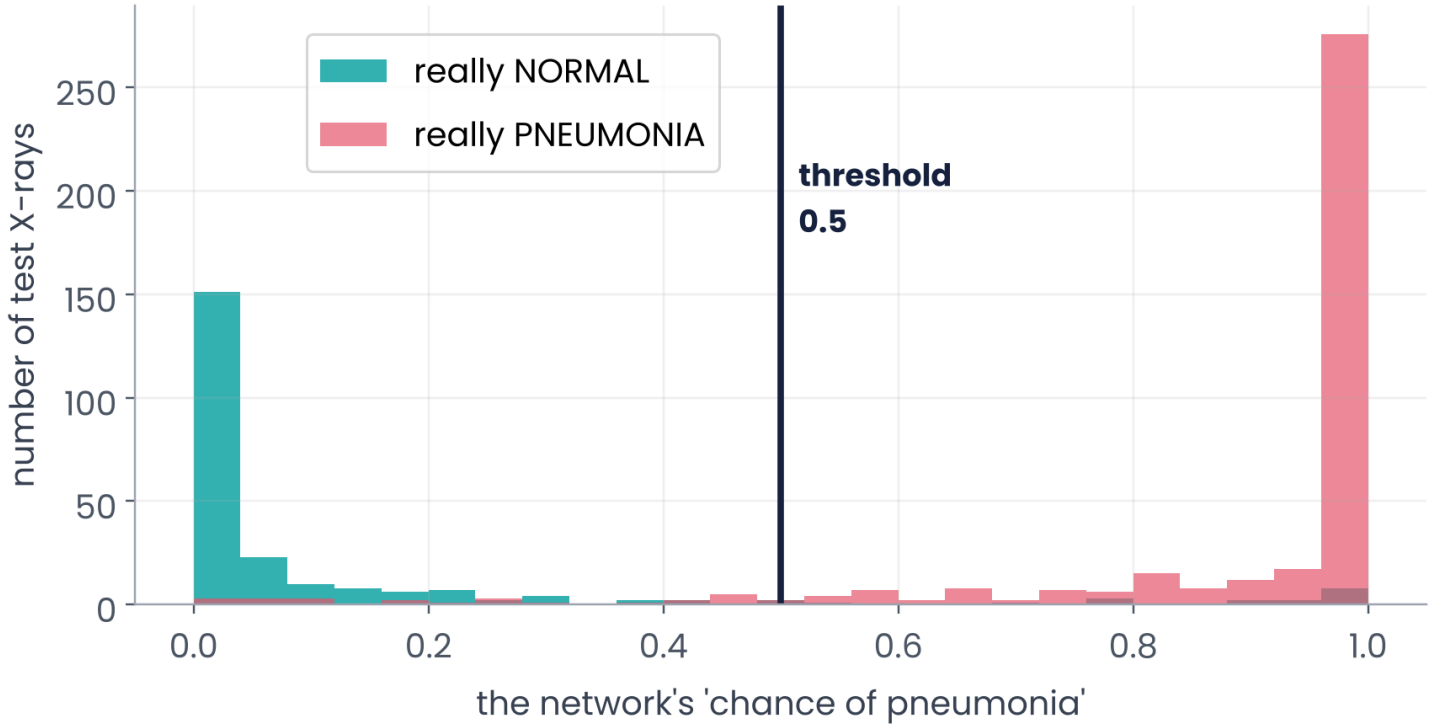
= 92.3%

A screening tool must have high sensitivity, because missing a sick child is worse than a false alarm.

We want both scores to be high. Each one tells only half of the story.

SIMULATION

The network gives a chance, not an answer



The threshold decides

For every X-ray, the network outputs a chance from 0 to 1.

Then **we** pick a **threshold**. Above it, we say "pneumonia." Below it, we say "normal."

Most X-rays are easy (near 0 or 1). The hard ones sit in the middle.

Who chooses the threshold? People, based on which mistake is worse.

Notice the few pink bars on the left: sick children the network felt sure were healthy.

SIMULATION

Moving the threshold

Threshold	Missed (sick)	False alarms	Sensitivity	Specificity	Accuracy
0.1	**8**	54	97.9%	76.9%	90.1%
0.22	**11**	33	97.2%	85.9%	92.9%
0.3	**16**	25	95.9%	89.3%	93.4%
0.5	**25**	18	93.6%	92.3%	93.1%
0.7	**47**	15	87.9%	93.6%	90.1%
0.9	**92**	12	76.4%	94.9%	83.3%

default 0.5 our choice 0.22

Lower threshold

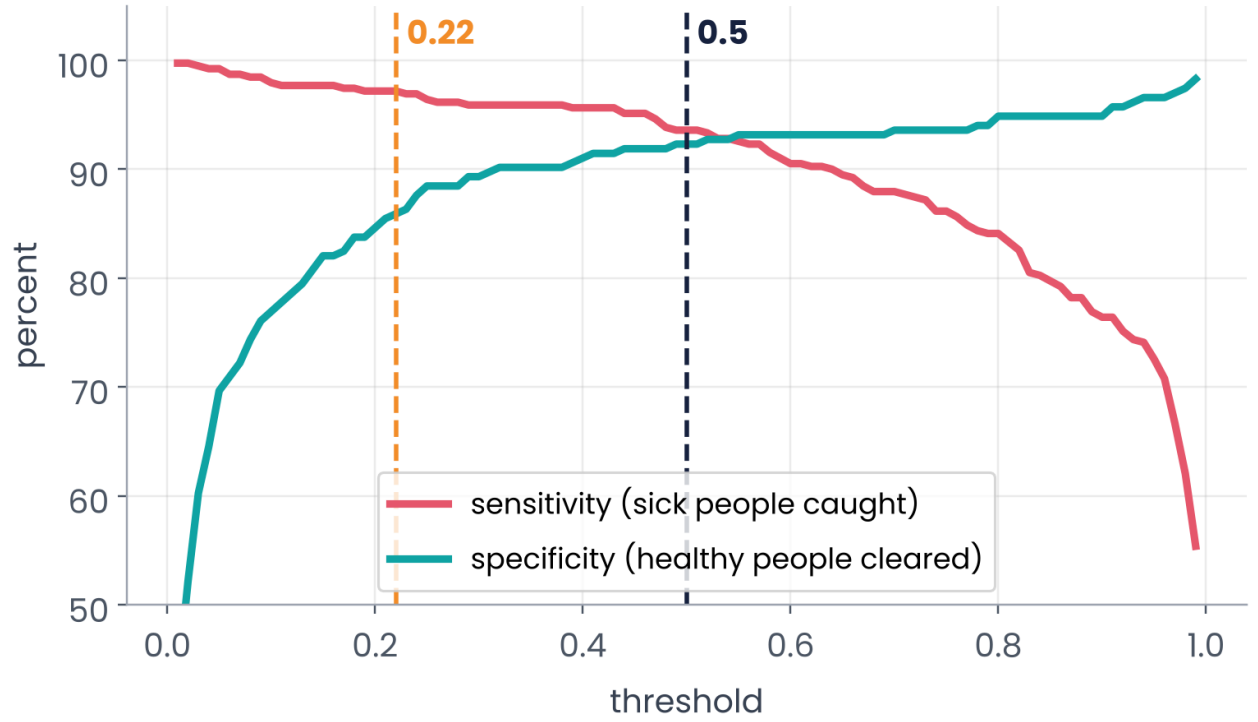
Catches more sick children, but creates more false alarms.

Higher threshold

Fewer false alarms, but more sick children are missed.

SIMULATION

Choosing a threshold, the careful way



Our rule

Catch at least **95%** of sick children. We found the threshold on the **validation set**: **0.22**. The test set stayed hidden.

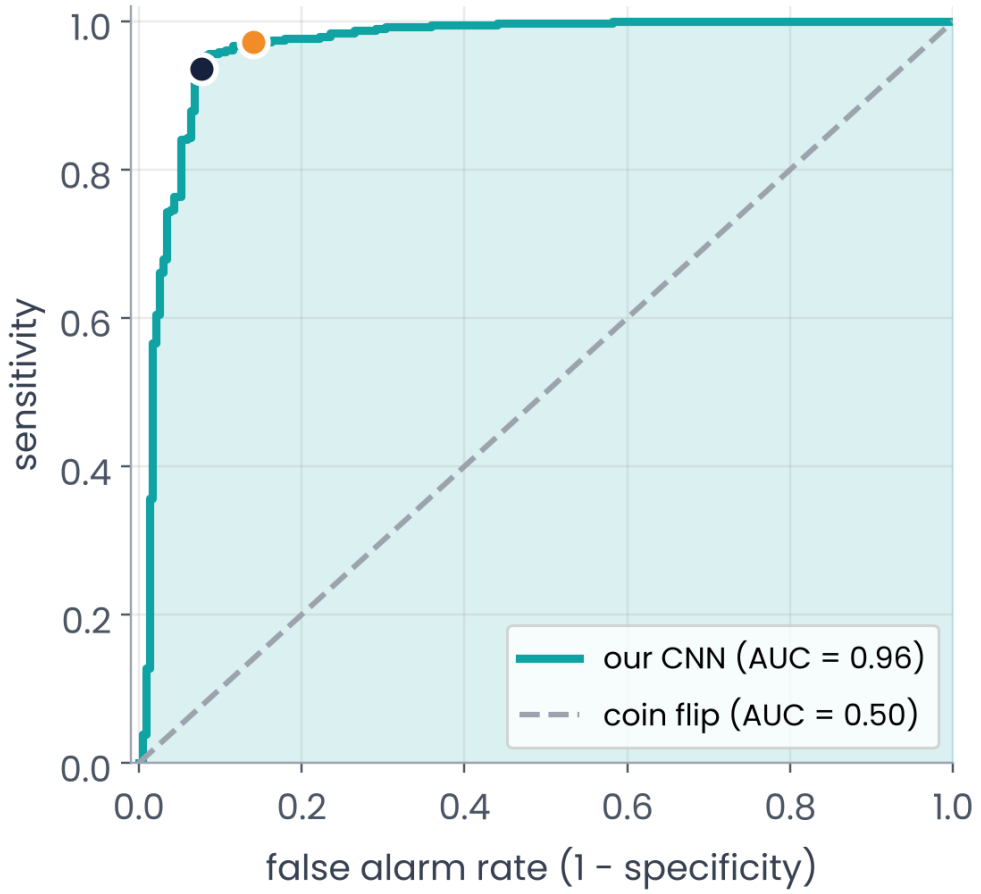
Result on the test set

Missed	25	→	11
False alarms	18	→	33
Sensitivity	93.6%	→	97.2%

We traded 15 extra false alarms for 14 fewer missed children. In a screening tool, that is usually worth it.

SIMULATION

The ROC curve and AUC



One curve, every threshold

Each point on the curve is one threshold. Up means more sick children caught. Right means more false alarms. The best curves hug the **top-left corner**.

AUC = 0.96

AUC is the **area under the curve**. Pick one sick and one healthy X-ray at random: about **96% of the time**, the network gives the sick one a higher chance.

AUC scale: 0.5 = coin flip · 0.7 = okay · 0.8 = good · 0.9+ = excellent. The dots mark thresholds 0.5 (navy) and our choice (orange).

A warning sign: practice vs. the real exam

Validation AUC

0.996

Test AUC

0.965

Why did the score drop?

In PneumoniaMNIST, the test X-rays come from a **different part** of the original collection than the training X-rays. New data is often a little different, and that makes it harder.

The real-world lesson

A model that works in one hospital may do **worse in another**, with different machines and patients. Medical AI must be tested on data from many places.

Always judge a model by data it has never seen, and expect real life to be harder still.

PART 6

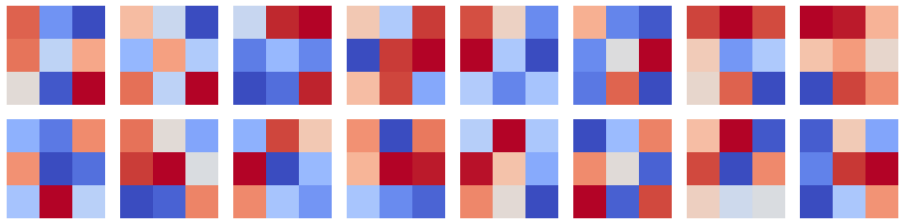
Opening the Black Box

A doctor explains a diagnosis. Can a CNN show us why it made its choice?



SIMULATION

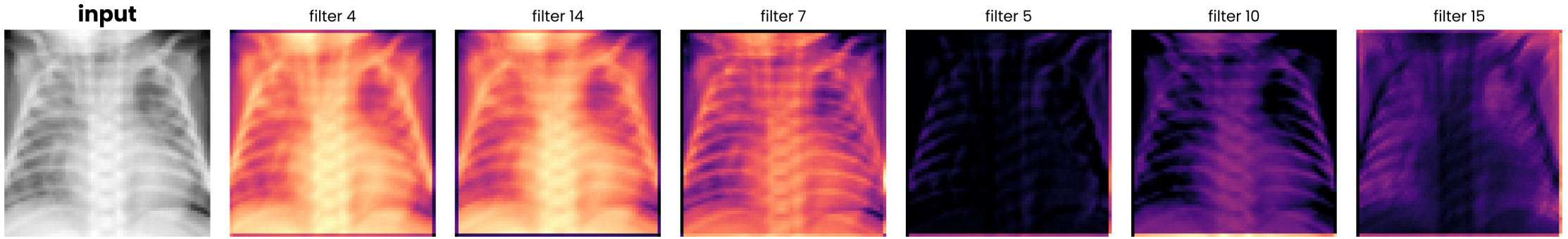
What did the first layer learn?



The 16 filters it learned (3 × 3). Red = positive, blue = negative.

Nobody designed these

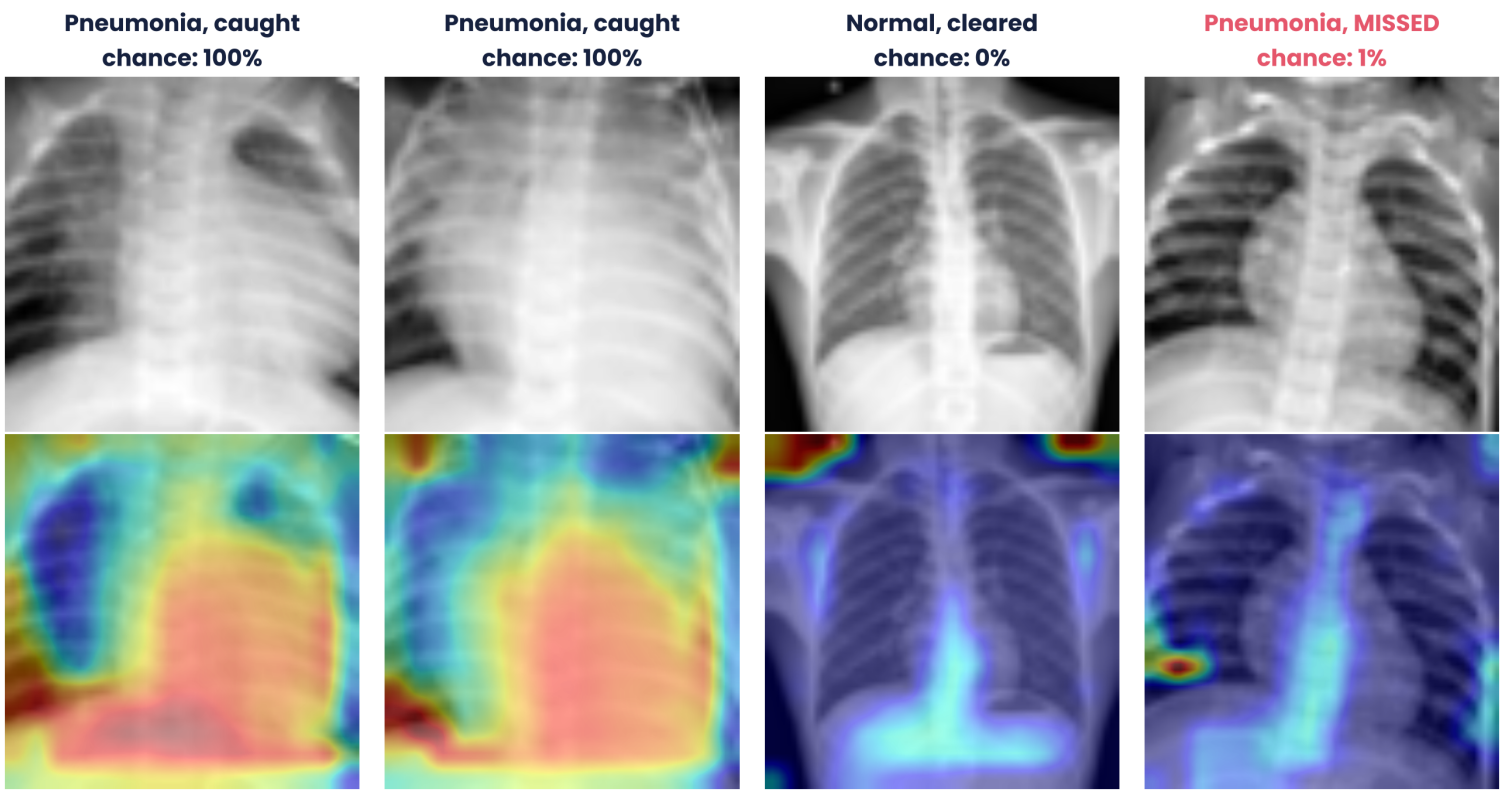
They started as random numbers. Training slowly shaped them into pattern detectors, just like the edge filters we drew by hand.



Feature maps: one X-ray seen through six learned filters. Some highlight bones, some the lung edges, some the dark air spaces.

SIMULATION

Grad-CAM: where is the network looking?



Heat map

Warm colors (red, yellow) show the areas that pushed the network **toward pneumonia**. Cool colors (blue) had little effect.

What we notice

For pneumonia, it looks at the **hazy white areas**. For the normal X-ray, the warm spots sit in the **corners, outside the lungs**. That is a warning sign worth checking (next slides).

SIMULATION

Learning from mistakes

missed pneumonia
chance 11%



missed pneumonia
chance 1%



missed pneumonia
chance 2%



false alarm
chance 97%



false alarm
chance 79%



false alarm
chance 97%



Missed cases

Some pneumonia X-rays look almost normal at 64 × 64 pixels. Small details are lost when the image is shrunk.

False alarms

Some normal X-rays are bright or blurry, which can look like hazy fluid to the network.

Extra objects

Look for wires, stickers, and markers. They are not part of the lungs, but they can confuse a network.

PART 6 · EXPLAIN

Danger: shortcut learning

What researchers have found

Some medical AI models scored well by looking at the **wrong things**: text markers printed on the X-ray, the type of machine used, or even which hospital the image came from.

1 Looks smart, isn't

The network finds an easy clue that happens to match the labels in the training data.

2 Breaks in new places

At a new hospital, the clue is gone, and the model suddenly fails.

3 How we catch it

Heat maps like Grad-CAM, testing on data from other hospitals, and doctors checking the results.

A high score is not proof. We must check why the model is right.

PART 7

Using Medical AI Responsibly

Health decisions affect real people. Here is what we must think about.



The limits of our model

Children only

All X-rays are of young children. It has never seen an adult chest.

One hospital

All images come from one place. Other machines and patients may look different.

Tiny images

At 64 × 64 pixels, many fine details are lost.

Only two answers

It cannot spot other illnesses, or tell which germ caused the pneumonia.

Labels can be wrong

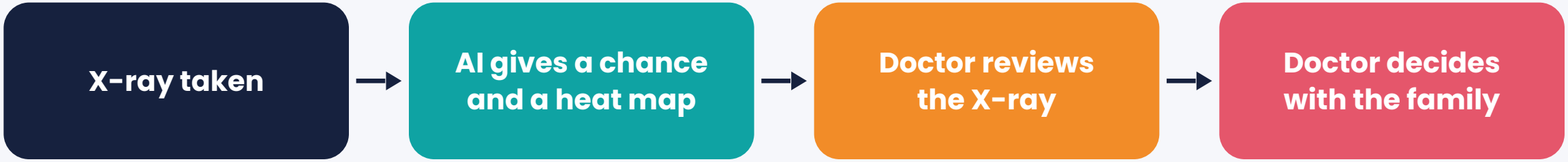
Even experts sometimes disagree, so some training labels may be mistakes.

Not approved

Real medical tools need years of testing and approval from health regulators.

Knowing the limits of a model is just as important as knowing its score.

Doctors and AI, working together



Is it fair?

Does it work equally well for every group of patients?

Is it safe?

Was it tested on new hospitals? What happens when it is wrong?

Is data private?

Were patient images shared with permission and with names removed?

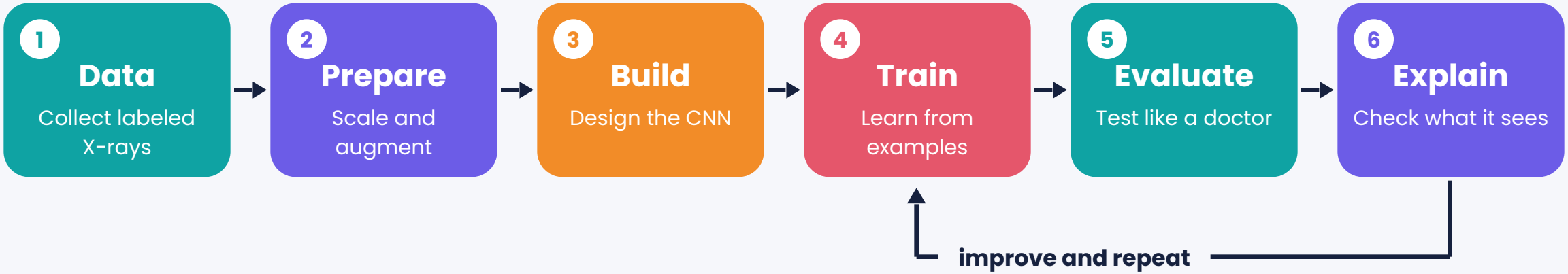
Who is responsible?

A trained person must always make, and own, the final decision.

Good medical AI builders ask these questions **before** a model ever reaches a patient.

WRAP-UP

The whole project in one picture



4,708
training X-rays

154,433
learned knobs

93.1%
test accuracy

97.2%
sensitivity at our threshold

0.96
test AUC

Good scores on one dataset are only the beginning. Real medical AI needs testing in many hospitals, doctors in the loop, and approval before it helps patients.

Quick check

1. Why do we keep the test set hidden until the very end?

2. A model that always says "pneumonia" gets 62.5% accuracy. Why is that a bad model?

3. What is the difference between sensitivity and specificity?

4. Why did we lower the threshold from 0.5?

5. Why do we NOT flip chest X-rays left to right during augmentation?

Quick check: answers

1. Why do we keep the test set hidden until the very end?

   So our final score is honest. It shows how the model does on X-rays it has never seen.

2. A model that always says "pneumonia" gets 62.5% accuracy. Why is that a bad model?

   It never looks at the X-ray. It sends every healthy child for treatment they do not need.

3. What is the difference between sensitivity and specificity?

   Sensitivity: how many sick people we catch. Specificity: how many healthy people we clear.

4. Why did we lower the threshold from 0.5?

   To catch more sick children, accepting a few more false alarms in exchange.

5. Why do we NOT flip chest X-rays left to right during augmentation?

   The heart is on the left side. A flipped chest would not look like a real X-ray.

KEEP EXPLORING

Try it yourself

1 Run the notebook

Every step in this lesson has a matching section. Train the CNN yourself in about 5 minutes on Google Colab.

2 Change the network

Remove a Conv block or turn off augmentation. Does the test AUC go up or down?

3 Pick your threshold

What threshold would you choose to catch 99% of sick children? How many false alarms does it cost?

Challenge: try another medical dataset

MedMNIST has more open datasets (CC BY 4.0) in the same format. In the notebook, change **one line** to train on **BreastMNIST** (breast ultrasound images, also a yes/no task). For a bigger challenge, try **BloodMNIST** (8 kinds of blood cells) or **OrganAMNIST** (11 organs from CT scans). These have many classes, so the output layer must change to one neuron per class with softmax.

WRAP-UP

Key takeaways

1. CNNs are made for pictures, so they are a natural fit for **medical images**.

2. Split the data into **training, validation, and test** sets, and keep the test set hidden.

3. **Class weights, augmentation, dropout,** and **early stopping** help a small model learn well.

4. Accuracy is not enough. Check the **confusion matrix, sensitivity,** and **specificity**.

5. The **threshold** is a human choice about which mistake is worse.

6. Explain the model, know its limits, and keep **doctors in charge**.

REFERENCES

Dataset credits

PneumoniaMNIST / MedMNIST v2

Yang, J., Shi, R., Wei, D., Liu, Z., Zhao, L., Ke, B., Pfister, H., & Ni, B. (2023). MedMNIST v2: A large-scale lightweight benchmark for 2D and 3D biomedical image classification. *Scientific Data*, 10, 41. License: CC BY 4.0. medmnist.com

Original chest X-ray images

Kermany, D. S., Goldbaum, M., et al. (2018). Identifying medical diagnoses and treatable diseases by image-based deep learning. *Cell*, 172(5), 1122–1131.

All results in this lesson come from one training run of the CNN described here. Your numbers may differ slightly.

Questions?

The best medical AI is built by people who care about both the math and the patients.

